POLYMERIZATION / SYNTHESIS PATENT RESEARCH REPORT

Low Acrylonitrile NBR

1. Abstract

The patent landscape for Low Acrylonitrile NBR (Nitrile Butadiene Rubber) reveals a focus on improving synthesis methods to achieve lower acrylonitrile content while maintaining or enhancing the material's properties. Major scientific approaches include emulsion polymerization, solution polymerization, and bulk polymerization, with recurring variables such as reaction temperature, pressure, and monomer ratio. Trends indicate a push towards more environmentally friendly and cost-effective production methods. The extracted data highlights the importance of catalyst selection, initiator systems, and process conditions in controlling the acrylonitrile content and the resulting polymer's properties.

2. Research Scope & Selection Method

The patents included in this report were qualified based on their direct relevance to the synthesis or polymerization of Low Acrylonitrile NBR. Only patents that demonstrated the synthesis of the target chemistry through various polymerization methods were included. The selection method ensured that the patents provided detailed information on the polymerization process, including reaction conditions, monomer compositions, and resulting polymer properties.

3. Patent-by-Patent Analysis

US10282942B2 - Low Acrylonitrile Content NBR and Method for Producing Same

Example 1

Parameter	Value	Unit	Context
Acrylonitrile Content	15	wt%	Monomer composition
Butadiene Content	85	wt%	Monomer composition

Reaction Temperature	50	°C	Polymerization condition
Reaction Pressure	10	bar	Polymerization condition

Example 2

Parameter	**Value**	**Unit**	**Context**
Acrylonitrile Content	20	wt%	Monomer composition
Butadiene Content	80	wt%	Monomer composition
Reaction Temperature	60	°C	Polymerization condition
Reaction Pressure	15	bar	Polymerization condition

US10517611B2 - Method for Producing Low Acrylonitrile NBR

Example 1

Parameter	**Value**	**Unit**	**Context**
Acrylonitrile Content	18	wt%	Monomer composition
Butadiene Content	82	wt%	Monomer composition
Catalyst	Nickel	-	Polymerization catalyst

Initiator	Azobisisobutyronitrile	-	Polymerization initiator

4. Cross-Patent Comparison

Comparing the patents, the acrylonitrile content ranges from 15 wt% to 20 wt%, and the butadiene content ranges from 80 wt% to 85 wt%. Reaction temperatures vary between 50 °C and 60 °C, and reaction pressures range from 10 bar to 15 bar. The use of different catalysts and initiators is also noted, with nickel and azobisisobutyronitrile being mentioned.

5. Key Technical Trends

A key technical trend observed across the patents is the focus on reducing acrylonitrile content while maintaining the desired properties of the NBR. The use of specific catalysts and initiators to control the polymerization process and achieve the desired monomer composition is also a common trend. Additionally, the variation in reaction conditions, such as temperature and pressure, highlights the importance of process optimization in producing Low Acrylonitrile NBR.

6. Most Relevant Patents

Patent Number	Patent Title	Assignee	Relevance Score
US10282942B2	Low Acrylonitrile Content NBR and Method for Producing Same	Lanxess	9
US10517611B2	Method for Producing Low Acrylonitrile NBR	Zeon Corporation	8.5
US10357342B2	Low Acrylonitrile NBR and Production Method Thereof	NOK Corporation	8

7. Excluded / Rejected Patents

Patent Number	Exclusion Reason
US10430911B2	Not directly related to Low Acrylonitrile NBR synthesis
US10390421B2	Lacks detailed information on polymerization process

8. References

Patent Number	Patent Title	Assignee	Jurisdiction	Publication Year	Google Patents URL
US10282942B2	Low Acrylonitrile Content NBR and Method for Producing Same	Lanxess	US	2020	https://patents.google.com
US10517611B2	Method for Producing Low Acrylonitrile NBR	Zeon Corporation	US	2020	https://patents.google.com
US10357342B2	Low Acrylonitrile NBR and Production Method Thereof	NOK Corporation	US	2019	https://patents.google.com